

MSBON Fraud-Sensitive Transcript Verification Project Proposal, Team Nexus

Team Information

Team Members: Suvi Lama, Chance Davis, Nishit Thapa, Sandesh Bhattarai, Kiran Silwal

Institution: The University Of Southern Mississippi

Date: February 27, 2026

Team Member Bios and Individual Pictures:

-Suvi Lama: Computer Science and Sociology major with experience in technology and society research.



-Kiran Silwal: Computer Science major with experience in app development and machine learning.



-Nishit Thapa: Computer Science major with experience in web development.



- Sandesh Bhattarai: Computer Science major with experience in app development.



- Chance Davis: Computer Science major with experience in system-level programming



Agency/Problem Area Our Project Addresses: Mississippi State Board of Nursing (MSBON)
Fraud-Sensitive Transcript Verification

1. Problem Understanding & Strategic Alignment

1.1 Agency Problem Description:

The MSBN currently reviews nursing school transcripts manually to confirm that applicants have completed an approved nursing education program. This process is time-consuming, inconsistent

across institutions, and vulnerable to human error, due to the wide variety of domestic nursing programs and varying state licensure requirements.

1.2 Why This Problem Matters:

Fraudulent or incomplete transcripts present a direct threat to public safety. Federal investigations like Operation Nightingale have demonstrated that bad actors can exploit gaps in manual review processes. A more reliable, auditable system is essential to ensure only fully qualified individuals receive licensure.

2. Proposed AI-Enabled Solution

2.1 Solution Overview:

An AI-assisted transcript verification tool that extracts and reviews key transcript elements, compares them against MSBN requirements, and flags potential anomalies for staff review, while preserving full human decision-making authority.

2.2 Key Features & Functionality:

- Automated extraction of coursework, degree conferral, and graduation indicators
- Rule-based comparison against MSBN licensure requirements
- Flagging of missing graduation confirmation, partial completions, formatting inconsistencies, and known fraud indicators
- Staff review interface for confirming, overriding, and annotating flags
- Audit-ready logs documenting rules applied, transcript sections reviewed, and rationale for each outcome
- Cross-referencing against accredited Mississippi nursing programs (ACEN, CCNE, etc.)

2.3 Innovation Statement:

This project uses transparent, rule-based AI to bring consistency and accountability to a high-stakes regulatory process without replacing the judgment of MSBN staff.

3. Impact Potential and Scalability

3.1 Expected Benefits:

- Reduced transcript review time for MSBN staff
- Standardized application of licensure criteria across all applicants
- Improved ability to detect and document potential fraud
- Fully explainable, auditable outputs that support appeals and compliance

3.2 Replicability & Scalability Potential:

This framework can be adapted for other state licensing boards that conduct credential verification, expanding its value across multiple agencies.

4. Feasibility & Technical Approach

4.1 Technical Feasibility:

The solution relies on document extraction (OCR for scanned transcripts), rule-based logic, and a staff-facing review dashboard. All outputs are advisory only; no automated licensure decisions are made.

4.2 Additional Data Requirements (list any data not noted in the scope that may be needed for successful PoC):

- Sample transcripts for PoC testing
- MSBN-defined education requirements, licensure rules, and SOPs
- List of currently accredited nursing programs in Mississippi
- Known fraud indicator criteria from MSBN staff

5. Project Scope Statement

5.1 In Scope:

AI-assisted transcript extraction and review, rule-based flagging of anomalies, staff review and override workflow, audit log generation, and accreditation cross-referencing for Mississippi nursing programs are in scope. Whereas, Automated licensure decisions, international transcript verification, integration with external systems, predictive risk scoring, production deployment, or public access are out of scope.

6. Timeline & Work Plan

Original Timeline & Work Plan:

Iterations	Deliverables	Planned Start Date
Week 1	● Kickoff ● Requirements confirmation ● Sample transcript selection	03/02/2026
Week 2	● Environment Setup ● Exploration of LLM models and techniques ● Data ingestion and transcript format analysis	03/09/2026
Weeks 3	● AI extraction and rule-based verification logic	03/26/2026(Spring Break in the week of March 16)
Week 4	● Human review workflow ● Source and explainability layer	03/30/2026

Week 5	• Data testing, review, and validation • Project Documentation	04/06/2026
Week 6	• Readout and demo with MSBON • Documentation	04/13/2026

7. Team Collaboration & Readiness

7.1 Team Roles & Responsibilities:

- Project Manager: Suvi
- Frontend Developer: Nishit
- Backend Developer: Sandesh
- Data/ML Engineer: Kiran
- QA/Testing & Documentation: Chance

7.2 Required Support:

- Sample transcript datasets from MSBN
- Transcript flagging rules and verification SOPs
- Access to accredited program lists
- Feedback from MSBN staff validators (Dr. Burks, Dr. Smith, Talisha Greenlaw)

8. Ethical, Legal, Privacy & Security Considerations

8.1 Risks or Concerns:

- False positives could unfairly delay legitimate applicants
- Transcripts contain sensitive personal and academic data
- OCR limitations may reduce accuracy for scanned or older documents

8.2 Mitigation Plan:

- All flags require human review before any action is taken
- Strict data handling and access controls throughout the PoC
- Only sample/test data used during development; no permanent transcript storage
- Transparent rule explanations included with every AI output

9. Additional Notes

The team will work closely with the Professor and MSBN staff to ensure outputs are trustworthy and easy to act on. The tool is designed to assist staff with every output fully explainable and auditable.